



An Image-based Method for the Morphological Analysis of Tendrils with 2D Piece-wise Clothoid Approximation Model

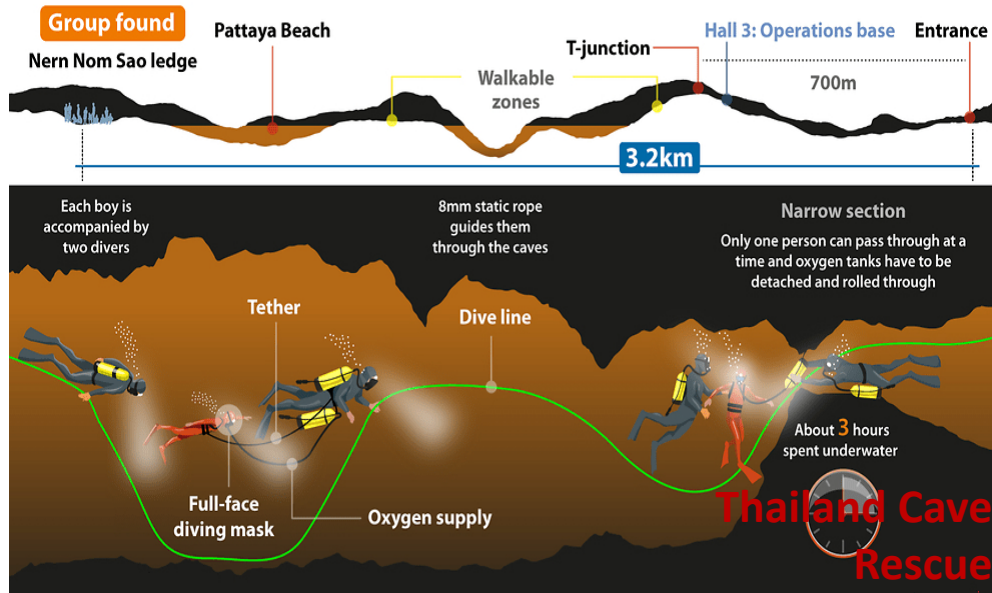
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Motivation

Exploration of unstructured environment difficult to reach by human beings



- Thin body
- Work on long distance
- Adaptable

- Filiform structure
- Moving by growing
- Anchorable



Proposed Method

Correctly describe and analyze the morphology of filiform structures

Linear property allows a smooth transition of curvatures

$\mathcal{K}(s) = k_1 s + k_2 \longrightarrow$ clothoid

$k_1 = 0 \longrightarrow$ circle

$k_1 = k_2 = 0 \longrightarrow$ straight line



Passionflower tendril



Clothoid curve

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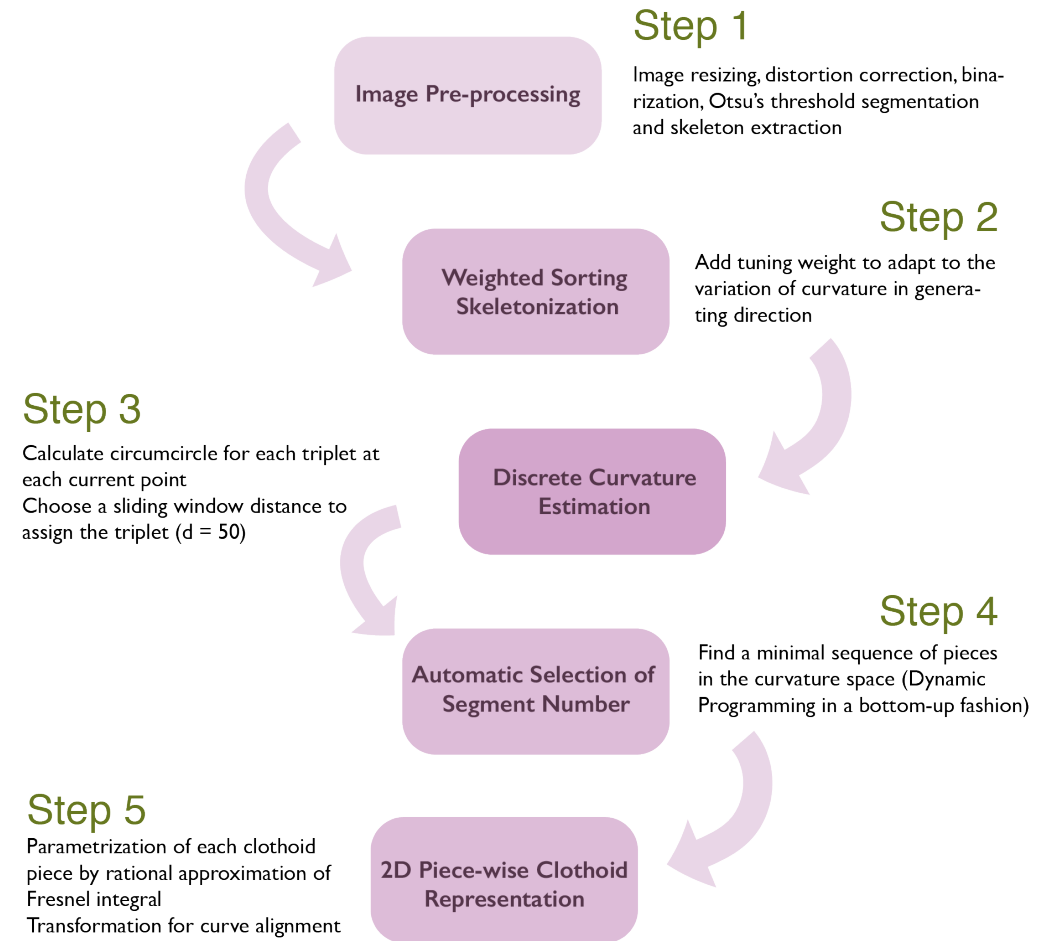
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Passionflower tendril



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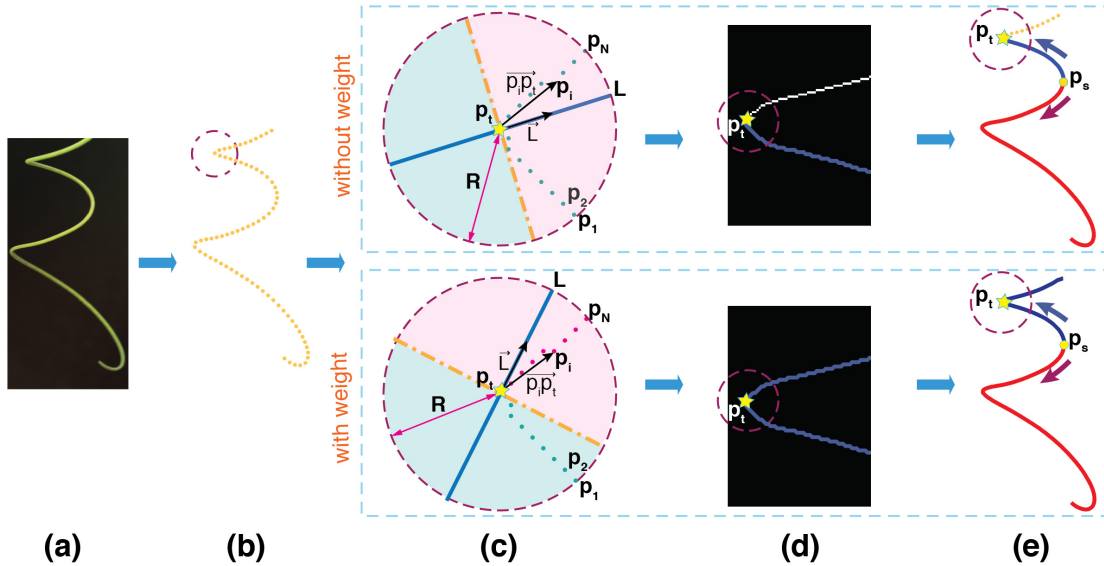


Flowchart of methodology

Proposed Method

Weighted Moving Least Squares Method – Step 2

(based on the previous work [Jie Fan et al. Robosoft 2020])



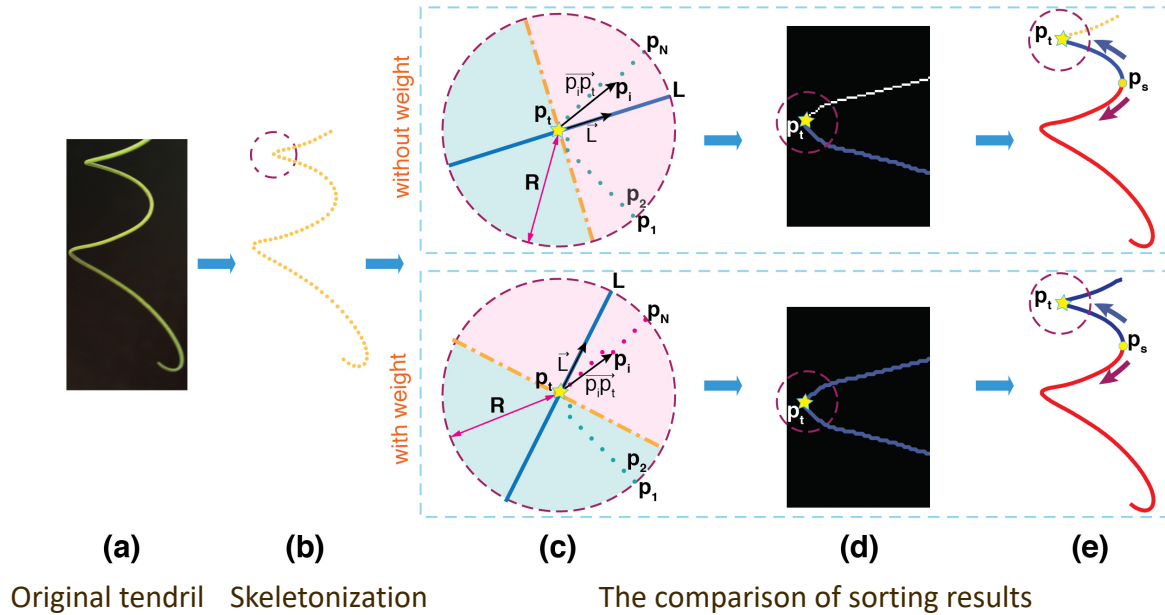
(a) Original tendril (b) Skeletonization

(c) The comparison of sorting results

Proposed Method

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Automatic Selection of Segment Number – Step 4

Dynamic Programming with an Up-sweep Approach

$$\mathcal{J}(p_i, p_j) = \min_{i < k < j} \{ \mathcal{J}(p_i, p_k) + \mathcal{J}(p_k, p_j), \underbrace{(\epsilon)}_{\text{penalty to generate a single segment}} + \mathcal{E}_{fit}(p_i, p_j) \}$$

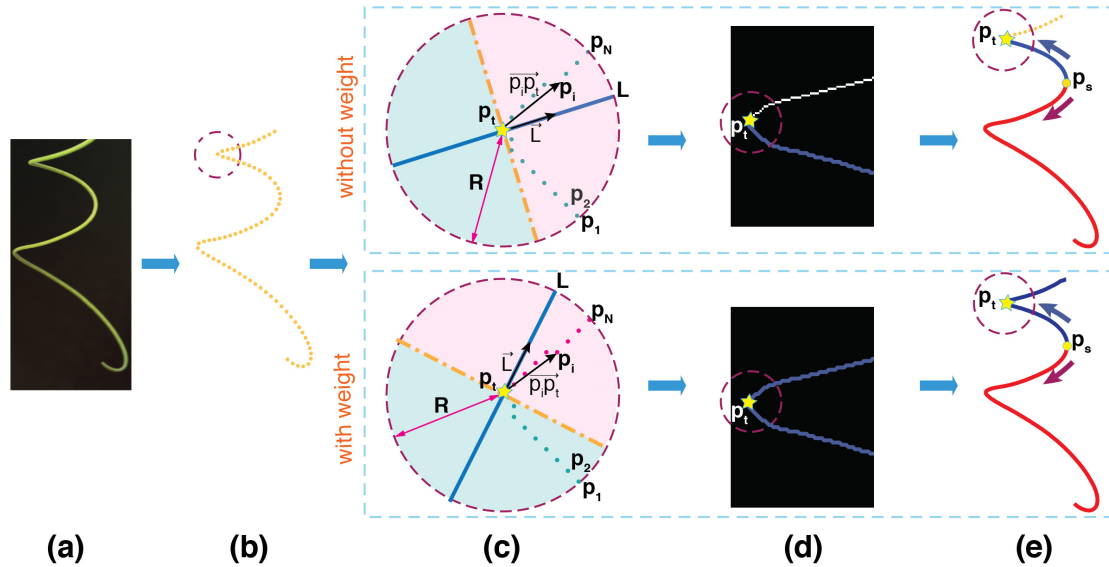
penalty to generate a single segment

$$\mathcal{E}_{fit}(p_i, p_j) = \min_{a_{ij}, b_{ij}} \left(\sum_{t=i}^j \left| \underbrace{a_{ij}}_{\text{slope}} \cdot \underbrace{s_t}_{\text{arclength}} + \underbrace{b_{ij}}_{\text{intercept}} - \underbrace{\kappa_t}_{\text{curvature}} \right|^2 \right)$$

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2D Piece-wise Clothoid Representation - Step 5

2D Clothoid Spiral with Fresnel Integrals

$$\begin{pmatrix} x \\ y \end{pmatrix} = \pi B \begin{pmatrix} C(l) \\ S(l) \end{pmatrix}$$

scaling factor

$$S(l) = \int_0^l \sin \frac{\pi}{2} \tau^2 d\tau \approx \frac{1}{2} - R(l) \cos \left(\frac{1}{2} \pi (A(l) - l^2) \right)$$

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penalty to generate a single segment

$$\mathcal{E}_{fit}(p_i, p_j) = \min_{a_{ij}, b_{ij}} \left(\sum_{t=i}^j \left(\overset{\text{slope}}{a_{ij}} \cdot \overset{\text{arclength}}{s_t} + \overset{\text{intercept}}{b_{ij}} - \overset{\text{curvature}}{\kappa_t} \right)^2 \right)$$

$$A(l) = \frac{1}{0.803l^3 + 1.886l^2 + 2.524l^2}$$

$$R(l) = \frac{0.506l + 1}{1.79l^2 + 2.054l + \sqrt{2}}$$

Approximation with 1.7×10^{-3} error on $R(l)$ and $A(l)$

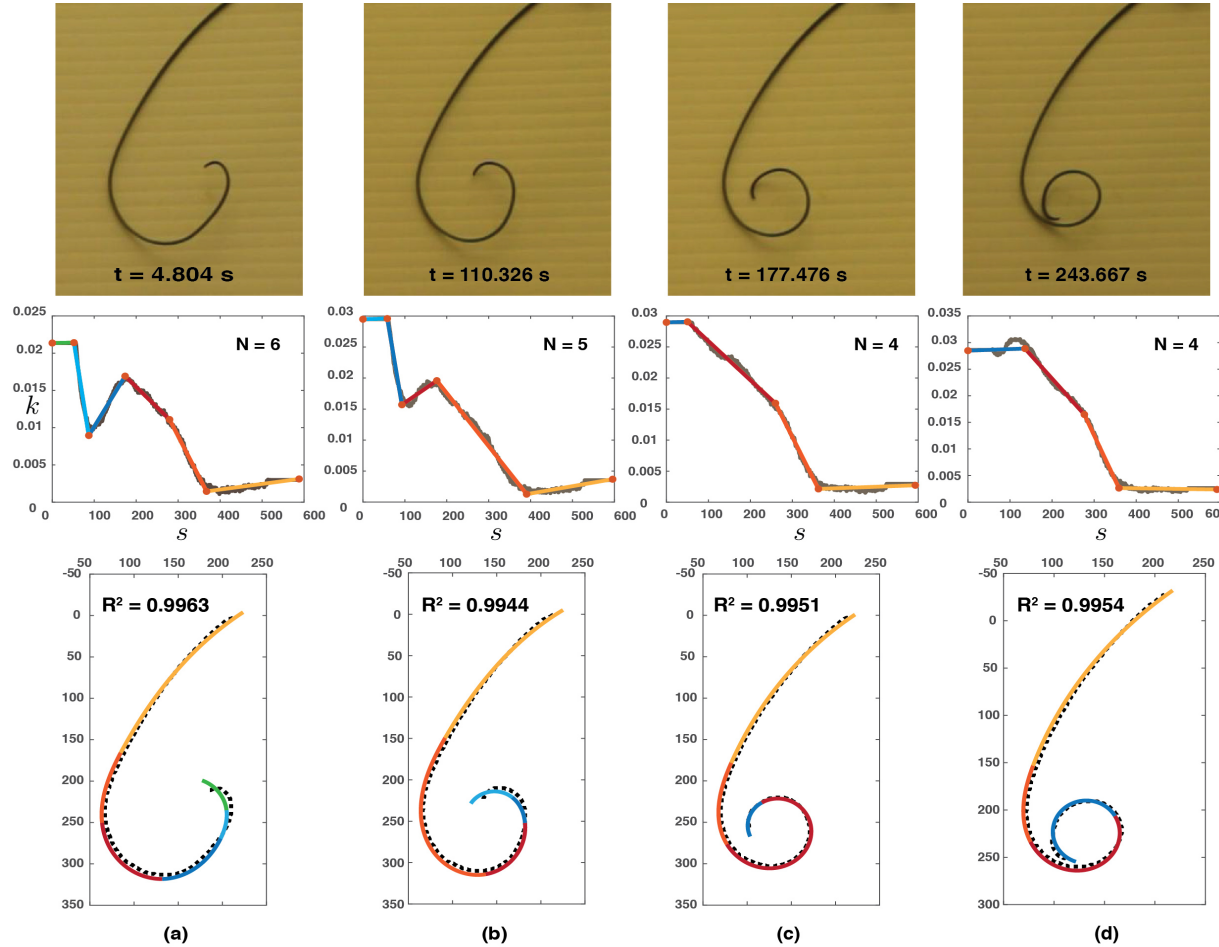
Results

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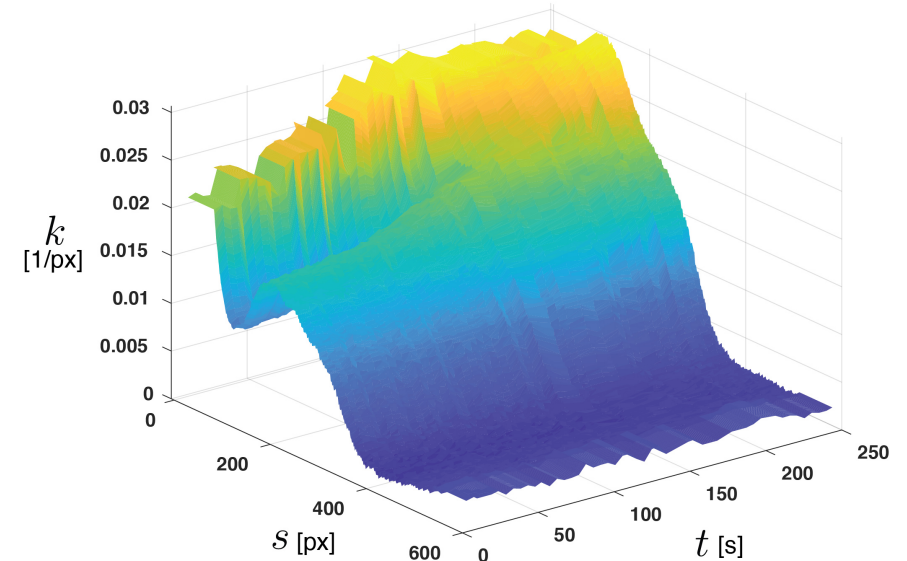
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We found that 4 to 6 segments were enough to represent curling shapes with high accuracy.

With this approach we can analyze the changes over time in the morphology of continuum adaptive structures.



Future Works

- Extend 3D model to better analyze the shape and behavior of tendrils
- Consider self-occlusion of tendrils during their growing evolution in 3D space
- Analyze the morphological evolution of natural tendrils during support grasping
- Use the extracted feature to guide the design of tendril-like soft robots
- Investigating and developing control strategies for continuum growing robots

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Thanks!

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